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Introduction

§ 1. Uncertainty Relations in the Relativistic Domain

The quantum theory presented in Vol. III of this course is non-relativistic in its very substance, and it fails for phenomena in which motion occurs at speeds no longer small next to the speed of light. One might at first imagine that a relativistic theory could be reached by generalizing the formalism of non-relativistic quantum mechanics in a more or less straightforward manner. A closer look shows, however, that a logically complete relativistic theory cannot be built without invoking fresh physical principles.

We begin by recalling several physical notions on which non-relativistic quantum mechanics rests (Vol. III, § 1). A fundamental part was played there, as we saw, by the concept of measurement — the process in which a quantum system interacts with a “classical object” (an “apparatus”) and thereby comes to possess definite values of particular dynamical variables (coordinates, velocities, and so forth). We saw, too, that quantum mechanics places severe limits on the extent to which different dynamical variables of an electron¹ can exist together. Thus the uncertainties Δq and Δp with which coordinate and momentum can coexist obey the relation $\Delta q \Delta p \sim \hbar$;² the more precisely either of these quantities is measured, the less precisely the other can be known at the same moment.

What matters, however, is that any single dynamical variable of the electron, on its own, admitted measurement of unlimited precision, and within a time interval that could be taken as short as desired. On this circumstance the whole of non-relativistic quantum mechanics depends in a fundamental way. It alone makes it possible to introduce the wave function, the central object of the theory’s formalism. The physical content of the wave function $\psi(q)$ is, after all, that its squared modulus gives the probability that a measurement performed at a given instant will yield one or another value of the electron’s coordinate. Evidently the very notion of such a probability presupposes that a coordinate measurement of unlimited precision and speed can be realized in principle; failing that, the notion would be empty of content and would forfeit its physical meaning.

The existence of a limiting speed (the speed of light c) leads to new limitations, of a fundamental kind, on the possibilities of measuring various physical quantities (L. D. Landau, R. Peierls, 1930).

In Vol. III, § 44 the relation

$$(v' - v) \Delta p \Delta t \sim \hbar \tag{1.1}$$

was obtained; it relates the uncertainty Δp of a momentum measurement on the electron to the length Δt of the measuring process itself, v and v' denoting the electron’s velocity before and after that process. It follows that momentum can be measured accurately within a short time (small Δp together with small Δt) only if the measurement itself

¹As in Vol. III, § 1, we say “electron” for brevity, understanding by it any quantum system.

²Ordinary units are used in this section.

alters the velocity strongly enough. In the non-relativistic theory this expresses the fact that a momentum measurement repeated after a short interval will not reproduce its result, but it places no restriction of principle on how accurate a single measurement of the momentum may be, the difference $v' - v$ being capable of unlimited increase.

The presence of a limiting speed, however, changes the state of affairs radically. The difference $v' - v$, like the velocities themselves, can now not exceed c (more precisely, $2c$). Replacing $v' - v$ in (1.1) by c , we obtain the relation

$$\Delta p \Delta t \sim \hbar/c, \quad (1.2)$$

which determines the best accuracy of momentum measurement attainable in principle for a given duration Δt of the measurement. Thus in the relativistic theory an arbitrarily accurate and arbitrarily rapid measurement of the momentum turns out to be impossible in principle. An exact measurement of the momentum ($\Delta p \rightarrow 0$) is possible only in the limit of an infinitely long duration of the measurement.

One has reason to expect that the measurability of the electron's coordinate as such is likewise affected. Within the mathematical formalism of the theory this appears as a conflict between an exact coordinate measurement and the statement that a free particle's energy is positive. As we shall see, the full set of eigenfunctions of a free particle's relativistic wave equation contains, besides the solutions whose time dependence is the "proper" one, solutions of "negative frequency" as well. These will in general appear also in the expansion of a wave packet describing an electron confined to a small region of space.

As will be shown, the "negative-frequency" wave functions are tied to the existence of antiparticles — the positrons. Their presence in the expansion of the wave packet reflects the creation of electron–positron pairs, unavoidable in the general case, during a measurement of the electron's coordinates. Since the emergence of these new particles cannot be controlled by the measuring process itself, coordinate measurement for the electron is drained of meaning.

In the rest frame of the electron the minimum error in a measurement of its coordinates is

$$\Delta q \sim \hbar/mc. \quad (1.3)$$

This value (the only one that dimensional considerations would permit) answers to a momentum uncertainty $\Delta p \sim mc$, which corresponds in turn to the minimum threshold energy needed to create a pair.

In a reference frame where the electron moves with energy ε , (1.3) is replaced by

$$\Delta q \sim c\hbar/\varepsilon. \quad (1.4)$$

In the extreme ultra-relativistic limit, in particular, energy and momentum are related by $\varepsilon \approx cp$, so that

$$\Delta q \sim \hbar/p, \quad (1.5)$$

the error Δq then coinciding with the particle's de Broglie wavelength.³

³What is meant here are measurements any outcome of which permits an inference about the state of

For photons the ultra-relativistic case always holds, so that expression (1.5) is valid. This means that it makes sense to speak of the coordinates of a photon only in those cases where the characteristic dimensions are large compared with the wavelength. But this is nothing other than the “classical” limiting case, corresponding to geometrical optics, in which one may speak of the propagation of light along definite trajectories — rays. In the quantum case, however, when the wavelength cannot be regarded as small, the concept of the coordinates of a photon becomes vacuous. We shall see later (see § 4) that in the mathematical formalism of the theory the non-measurability of the photon’s coordinates already shows itself in the impossibility of forming from its wave function a quantity that could play the part of a probability density satisfying the requirements which relativistic invariance demands.

On the basis of all that has been said, it is natural to suppose that the future theory will renounce altogether the consideration of the time development of the processes of interaction of particles. It will show that in these processes no exactly definable characteristics exist (even within the limits of the usual quantum-mechanical accuracy), so that the description of a process in time will turn out to be just as illusory as classical trajectories proved to be in non-relativistic quantum mechanics. The only observable quantities will be the characteristics (momenta, polarizations) of free particles — the initial particles that enter into the interaction and the final particles that have arisen as a result of the process (*L. D. Landau, R. Peierls, 1930*).

The characteristic formulation of a problem in relativistic quantum theory consists in the determination of the probability amplitudes of transitions connecting given initial and final states (that is, at $t \rightarrow \mp\infty$) of a system of particles. The totality of the transition amplitudes between all possible states constitutes the *scattering matrix*, or *S-matrix*. This matrix will be the carrier of all the information about the processes of interaction of particles that possesses an observable physical meaning (*W. Heisenberg, 1938*).

At the present time a complete, logically closed relativistic quantum theory does not yet exist. We shall see that the existing theory introduces new physical aspects into the manner of describing the states of particles, which acquires certain features of field theory (see § 10). The theory is constructed, however, to a considerable degree on the model of, and with the help of the concepts of, ordinary quantum mechanics. This way of building the theory has led to success in the domain of quantum electrodynamics. The absence of complete logical closure shows itself in this theory in the existence of divergent expressions when its mathematical apparatus is applied directly; but for the removal of these divergences there exist entirely unambiguous procedures. Nevertheless these procedures retain to a considerable degree the character of semi-empirical recipes, and our confidence in the correctness of the results obtained in this way rests, in the end, on their excellent agreement with experiment, and not on the internal consistency and logical coherence of the fundamental principles of the theory.

the electron; we thus set aside coordinate measurements by collision, in which over the time of observation the result occurs with a probability short of unity. In such a case a deflection of the particle does allow the electron’s position to be inferred, but from the absence of a deflection nothing at all can be concluded.

Photon

§2. Quantization of the Free Electromagnetic Field

In order to treat the electromagnetic field as a quantum object, it is convenient to start from a classical description in which the field is characterized by a discrete, albeit infinite, set of variables; such a description will allow one to apply the standard formalism of quantum mechanics directly. A description by means of potentials specified at every point of space is, in essence, a description in terms of a continuous set of variables.

Let $\mathbf{A}(\mathbf{r}, t)$ be the vector potential of the free electromagnetic field, satisfying the “transversality condition”

$$\nabla \cdot \mathbf{A} = 0. \quad (2.1)$$

The scalar potential then vanishes, $\Phi = 0$, and the fields $\mathbf{E}$ and $\mathbf{H}$ are

$$\mathbf{E} = -\dot{\mathbf{A}}, \quad \mathbf{H} = \nabla \times \mathbf{A}. \quad (2.2)$$

Maxwell’s equations reduce to the wave equation for $\mathbf{A}$:

$$\Delta \mathbf{A} - \frac{\partial^2 \mathbf{A}}{\partial t^2} = 0. \quad (2.3)$$

As is well known (see Vol. II, § 52), in classical electrodynamics the passage to a description in terms of a discrete set of variables is effected by considering the field inside a large but finite volume V .¹ Let us recall, omitting the details of the calculation, how this is done.

A field in a finite volume can be expanded in travelling plane waves, so that its potential takes the form of a series

$$\mathbf{A} = \sum_{\mathbf{k}} (\mathbf{a}_{\mathbf{k}} e^{i\mathbf{k}\mathbf{r}} + \mathbf{a}_{\mathbf{k}}^* e^{-i\mathbf{k}\mathbf{r}}), \quad (2.4)$$

where the coefficients $\mathbf{a}_{\mathbf{k}}$ depend on time according to

$$\mathbf{a}_{\mathbf{k}} \sim e^{-i\omega t}, \quad \omega = |\mathbf{k}|. \quad (2.5)$$

By virtue of condition (2.1) the complex vectors $\mathbf{a}_{\mathbf{k}}$ are orthogonal to the corresponding wave vectors: $\mathbf{a}_{\mathbf{k}} \mathbf{k} = 0$.

The summation in (2.4) runs over an infinite discrete set of values of the wave vector (its three components k_x, k_y, k_z). The transition to an integration over a continuous distribution is made with the aid of the expression

$$\frac{d^3 k}{(2\pi)^3}$$

¹To avoid burdening formulae with superfluous factors, we set $V = 1$.

for the number of allowed values of $\mathbf{k}$ falling within a volume element $d^3k = dk_x dk_y dk_z$ of $\mathbf{k}$ -space.

Once the vectors $\mathbf{a}_\mathbf{k}$ are given, the field inside the volume in question is completely determined. One may therefore view these quantities as a discrete set of classical “field variables”. In order to find the passage to a quantum theory, however, a further transformation of these quantities is needed, one that brings the field equations into a form analogous to the canonical (Hamiltonian) equations of classical mechanics. The canonical variables of the field are introduced by means of

$$\mathbf{Q}_\mathbf{k} = \frac{1}{\sqrt{4\pi}}(\mathbf{a}_\mathbf{k} + \mathbf{a}_\mathbf{k}^*), \quad \mathbf{P}_\mathbf{k} = -\frac{i\omega}{\sqrt{4\pi}}(\mathbf{a}_\mathbf{k} - \mathbf{a}_\mathbf{k}^*) = \dot{\mathbf{Q}}_\mathbf{k} \quad (2.6)$$

(they are evidently real). The vector potential is expressed through the canonical variables as

$$\mathbf{A} = \sqrt{4\pi} \sum_{\mathbf{k}} \left(\mathbf{Q}_\mathbf{k} \cos \mathbf{k}\mathbf{r} - \frac{1}{\omega} \mathbf{P}_\mathbf{k} \sin \mathbf{k}\mathbf{r} \right). \quad (2.7)$$

To find the Hamiltonian H one must evaluate the total field energy

$$\frac{1}{8\pi} \int (\mathbf{E}^2 + \mathbf{H}^2) d^3x,$$

expressed in terms of the quantities $\mathbf{Q}_\mathbf{k}$, $\mathbf{P}_\mathbf{k}$. Substituting the expansion (2.7) for $\mathbf{A}$, computing $\mathbf{E}$ and $\mathbf{H}$ by means of (2.2), and integrating, one obtains

$$H = \frac{1}{2} \sum_{\mathbf{k}} (\mathbf{P}_\mathbf{k}^2 + \omega^2 \mathbf{Q}_\mathbf{k}^2).$$

Each of the vectors $\mathbf{P}_\mathbf{k}$ and $\mathbf{Q}_\mathbf{k}$ is perpendicular to the wave vector $\mathbf{k}$ and therefore has two independent components. The direction of these vectors fixes the polarization direction of the corresponding wave. Denoting the two components of $\mathbf{Q}_\mathbf{k}$, $\mathbf{P}_\mathbf{k}$ (in the plane perpendicular to $\mathbf{k}$) by $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$ ($\alpha = 1, 2$), we rewrite the Hamiltonian as

$$H = \sum_{\mathbf{k}\alpha} \frac{1}{2} (P_{\mathbf{k}\alpha}^2 + \omega^2 Q_{\mathbf{k}\alpha}^2). \quad (2.8)$$

The Hamiltonian has thus broken up into a sum of independent terms, each involving just one pair $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$. Every term answers to a travelling wave of a given wave vector and polarization and reproduces the structure of a single harmonic-oscillator Hamiltonian. One accordingly speaks of this decomposition as the expansion of the field in *oscillators*.

We now turn to the quantization of the free electromagnetic field. The method of classical description just set out makes the route to quantum theory obvious. The canonical variables — the generalized coordinates $Q_{\mathbf{k}\alpha}$ and generalized momenta $P_{\mathbf{k}\alpha}$ — are now to be treated as operators obeying the commutation rule

$$\hat{P}_{\mathbf{k}\alpha} \hat{Q}_{\mathbf{k}\alpha} - \hat{Q}_{\mathbf{k}\alpha} \hat{P}_{\mathbf{k}\alpha} = -i \quad (2.9)$$

(operators with different $\mathbf{k}\alpha$ all commute with one another). Together with them the potential $\mathbf{A}$ and, by (2.2), the field strengths $\mathbf{E}$ and $\mathbf{H}$ also become (Hermitian) operators.

A consistent determination of the field Hamiltonian requires evaluation of the integral

$$\hat{H} = \frac{1}{8\pi} \int (\hat{\mathbf{E}}^2 + \hat{\mathbf{H}}^2) d^3x, \quad (2.10)$$

in which $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ are expressed through the operators $\hat{P}_{\mathbf{k}\alpha}$, $\hat{Q}_{\mathbf{k}\alpha}$. In practice, however, the non-commutativity of the latter does not make itself felt here, since the products $Q_{\mathbf{k}\alpha}P_{\mathbf{k}\alpha}$ enter with the factor $\cos \mathbf{k}\mathbf{r} \cdot \sin \mathbf{k}\mathbf{r}$, which vanishes upon integration over the full volume. Consequently the Hamiltonian takes the form

$$\hat{H} = \sum_{\mathbf{k}\alpha} \frac{1}{2} (\hat{P}_{\mathbf{k}\alpha}^2 + \omega^2 \hat{Q}_{\mathbf{k}\alpha}^2), \quad (2.11)$$

exactly reproducing the classical Hamiltonian, as one would naturally expect.

Finding the eigenvalues of this Hamiltonian requires no special computation, since the problem reduces to the familiar one of the energy levels of linear oscillators (see Vol. III, § 23). We can therefore write down at once the energy levels of the field:

$$E = \sum_{\mathbf{k}\alpha} \left(N_{\mathbf{k}\alpha} + \frac{1}{2} \right) \omega, \quad (2.12)$$

where $N_{\mathbf{k}\alpha}$ are non-negative integers.

Discussion of this formula will be taken up in the next section; here we write out the matrix elements of the quantities $Q_{\mathbf{k}\alpha}$, which can be obtained directly from the known formulae for the matrix elements of the oscillator coordinate (see Vol. III, § 23). The non-vanishing matrix elements are

$$\langle N_{\mathbf{k}\alpha} | Q_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} - 1 \rangle = \sqrt{\frac{N_{\mathbf{k}\alpha}}{2\omega}}. \quad (2.13)$$

The matrix elements of $P_{\mathbf{k}\alpha} = \dot{Q}_{\mathbf{k}\alpha}$ differ from those of $Q_{\mathbf{k}\alpha}$ only by a factor $\pm i\omega$.

For subsequent calculations it will, however, be more convenient to work not with $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$ themselves but with their linear combinations $\omega Q_{\mathbf{k}\alpha} \pm iP_{\mathbf{k}\alpha}$, which have matrix elements only for the transitions $N_{\mathbf{k}\alpha} \rightarrow N_{\mathbf{k}\alpha} \pm 1$. Accordingly we introduce the operators

$$\hat{c}_{\mathbf{k}\alpha} = \frac{1}{\sqrt{2\omega}} (\omega \hat{Q}_{\mathbf{k}\alpha} + i \hat{P}_{\mathbf{k}\alpha}), \quad \hat{c}_{\mathbf{k}\alpha}^+ = \frac{1}{\sqrt{2\omega}} (\omega \hat{Q}_{\mathbf{k}\alpha} - i \hat{P}_{\mathbf{k}\alpha}) \quad (2.14)$$

(the classical quantities $c_{\mathbf{k}\alpha}$, $c_{\mathbf{k}\alpha}^*$ coincide, up to a factor $\sqrt{2\pi/\omega}$, with the coefficients $a_{\mathbf{k}\alpha}$, $a_{\mathbf{k}\alpha}^*$ in the expansion (2.4)).

The matrix elements of these operators are

$$\langle N_{\mathbf{k}\alpha} - 1 | \hat{c}_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} \rangle = \langle N_{\mathbf{k}\alpha} | \hat{c}_{\mathbf{k}\alpha}^+ | N_{\mathbf{k}\alpha} - 1 \rangle = \sqrt{N_{\mathbf{k}\alpha}}. \quad (2.15)$$

The commutation rule between $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}_{\mathbf{k}\alpha}^+$ follows from the definition (2.14) and the rule (2.9):

$$\hat{c}_{\mathbf{k}\alpha} \hat{c}_{\mathbf{k}\alpha}^+ - \hat{c}_{\mathbf{k}\alpha}^+ \hat{c}_{\mathbf{k}\alpha} = 1. \quad (2.16)$$

Returning to the vector potential, we adopt an expansion of the same type as (2.4), but with the coefficients now promoted to operators:

$$\hat{\mathbf{A}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} \mathbf{A}_{\mathbf{k}\alpha} + \hat{c}_{\mathbf{k}\alpha}^+ \mathbf{A}_{\mathbf{k}\alpha}^*), \quad (2.17)$$

where

$$\mathbf{A}_{\mathbf{k}\alpha} = \sqrt{4\pi} \frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}} e^{i\mathbf{k}\mathbf{r}}. \quad (2.18)$$

Here $\mathbf{e}^{(\alpha)}$ denotes a unit vector specifying the polarization direction of the oscillator; the vectors $\mathbf{e}^{(\alpha)}$ are perpendicular to the wave vector $\mathbf{k}$, and for each $\mathbf{k}$ there are two independent polarizations.

Analogously, for the operators $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ we write

$$\hat{\mathbf{E}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} \mathbf{E}_{\mathbf{k}\alpha} + \hat{c}_{\mathbf{k}\alpha}^+ \mathbf{E}_{\mathbf{k}\alpha}^*), \quad \hat{\mathbf{H}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} \mathbf{H}_{\mathbf{k}\alpha} + \hat{c}_{\mathbf{k}\alpha}^+ \mathbf{H}_{\mathbf{k}\alpha}^*), \quad (2.19)$$

with

$$\mathbf{E}_{\mathbf{k}\alpha} = i\omega \mathbf{A}_{\mathbf{k}\alpha}, \quad \mathbf{H}_{\mathbf{k}\alpha} = \mathbf{n} \times \mathbf{E}_{\mathbf{k}\alpha} \quad (\mathbf{n} = \mathbf{k}/\omega). \quad (2.20)$$

The vectors $\mathbf{A}_{\mathbf{k}\alpha}$ are mutually orthogonal in the sense that

$$\int \mathbf{A}_{\mathbf{k}\alpha} \mathbf{A}_{\mathbf{k}'\alpha'}^* d^3x = \frac{2\pi}{\omega} \delta_{\alpha\alpha'} \delta_{\mathbf{k}\mathbf{k}'}. \quad (2.21)$$

To see this, note that when $\mathbf{A}_{\mathbf{k}\alpha}$ and $\mathbf{A}_{\mathbf{k}'\alpha'}^*$ carry different wave vectors their product involves $e^{i(\mathbf{k}-\mathbf{k}')\mathbf{r}}$, which integrates to zero over the volume; when only the polarizations differ, the orthogonality $\mathbf{e}^{(\alpha)} \mathbf{e}^{(\alpha')*} = 0$ of the two independent polarization directions provides the required vanishing. Analogous relations hold for the vectors $\mathbf{E}_{\mathbf{k}\alpha}$, $\mathbf{H}_{\mathbf{k}\alpha}$. Their normalization is conveniently written as

$$\frac{1}{4\pi} \int (\mathbf{E}_{\mathbf{k}\alpha} \mathbf{E}_{\mathbf{k}'\alpha'}^* + \mathbf{H}_{\mathbf{k}\alpha} \mathbf{H}_{\mathbf{k}'\alpha'}^*) d^3x = \omega \delta_{\mathbf{k}\mathbf{k}'} \delta_{\alpha\alpha'}. \quad (2.22)$$

Substituting the operators (2.19) into (2.10) and integrating with the help of (2.22), we get the field Hamiltonian expressed through the operators $\hat{c}$, $\hat{c}^+$:

$$\hat{H} = \sum_{\mathbf{k}\alpha} \frac{1}{2} \omega (\hat{c}_{\mathbf{k}\alpha} \hat{c}_{\mathbf{k}\alpha}^+ + \hat{c}_{\mathbf{k}\alpha}^+ \hat{c}_{\mathbf{k}\alpha}). \quad (2.23)$$

In the representation under discussion (with the matrix elements of $\hat{c}$, $\hat{c}^+$ given by (2.15)) this operator is diagonal, and its eigenvalues coincide, naturally, with (2.12).

In classical theory the field momentum is defined by the integral

$$\mathbf{P} = \frac{1}{4\pi} \int \mathbf{E} \times \mathbf{H} d^3x.$$

On passing to the quantum theory by replacing $\mathbf{E}$ and $\mathbf{H}$ with the operators (2.19), one readily finds

$$\hat{\mathbf{P}} = \sum_{\mathbf{k}\alpha} \frac{1}{2} (\hat{P}_{\mathbf{k}\alpha}^2 + \omega^2 \hat{Q}_{\mathbf{k}\alpha}^2) \mathbf{n} \quad (2.24)$$

— which mirrors the familiar classical link between energy and momentum of plane waves. This operator has eigenvalues

$$\mathbf{P} = \sum_{\mathbf{k}\alpha} \mathbf{k} \left(N_{\mathbf{k}\alpha} + \frac{1}{2} \right). \quad (2.25)$$

The operator representation realized by the matrix elements (2.15) is called the “occupation-number representation”: it amounts to characterizing the state of the system (the field) through the quantum numbers $N_{\mathbf{k}\alpha}$ (*occupation numbers*). Within this representation the field operators (2.19) (and hence the Hamiltonian (2.11)) act on the system’s wave function, written as a function of the numbers $N_{\mathbf{k}\alpha}$; we denote it $\Phi(N_{\mathbf{k}\alpha}, t)$. The field operators (2.19) do not depend on time explicitly. This corresponds to the Schrödinger representation of operators, familiar from non-relativistic quantum mechanics. What does depend on time is the state of the system, $\Phi(N_{\mathbf{k}\alpha}, t)$, its time dependence being governed by the Schrödinger equation

$$i \frac{\partial \Phi}{\partial t} = \hat{H} \Phi.$$

This description of the field is relativistically invariant in essence, since it rests on the invariant Maxwell equations. Yet the invariance is not exhibited explicitly — above all because the spatial coordinates and the time enter the description in a highly asymmetric way.

Within a relativistic framework it is desirable to give the description a more overtly invariant appearance. For this purpose one uses the so-called Heisenberg representation, wherein the explicit dependence on time is shifted onto the operators (see Vol. III, § 13). Time and spatial coordinates then appear symmetrically in the field-operator expressions, while the system state Φ depends only on the occupation numbers.

For the operator $\hat{\mathbf{A}}$ the passage to the Heisenberg representation amounts to replacing, in each term of the sum in (2.17), (2.18), the factor $e^{i\mathbf{k}\mathbf{r}}$ by $e^{i(\mathbf{k}\mathbf{r} - \omega t)}$; that is, one is to understand by $\mathbf{A}_{\mathbf{k}\alpha}$ the time-dependent functions

$$\mathbf{A}_{\mathbf{k}\alpha} = \sqrt{4\pi} \frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}} e^{-i(\omega t - \mathbf{k}\mathbf{r})}. \quad (2.26)$$

This is easily verified by noting that the matrix element of a Heisenberg operator for the transition $i \rightarrow f$ must contain the factor $\exp\{-i(E_i - E_f)t\}$, where E_i and E_f are the energies of the initial and final states (see Vol. III, § 13). For a transition in which $N_{\mathbf{k}}$ decreases or increases by one, this factor reduces to $e^{-i\omega t}$ or $e^{i\omega t}$, respectively. The indicated replacement satisfies this requirement.

Henceforth (in treating both the electromagnetic field and the fields of particles) we shall always understand the Heisenberg representation of operators.

§3. Photons

Let us now turn to a discussion of the quantization formulae derived above.

To begin with, formula (2.12) for the energy of the field exposes the following difficulty. At the lowest energy level, all oscillator quantum numbers $N_{\mathbf{k}\alpha}$ vanish (this

state is known as the *electromagnetic field vacuum*). Yet even in this state every oscillator retains a nonzero “zero-point energy” $\omega/2$. Summing over the entire infinite set of oscillators then produces an infinite result. One is thus confronted with one of the “divergences” stemming from the absence of full logical self-consistency in the existing theory.

As long as one is concerned only with the eigenvalues of the field energy, this difficulty can be eliminated by simply subtracting away the zero-point oscillation energy, i.e. by writing for the energy and momentum of the field²

$$E = \sum_{\mathbf{k}\alpha} N_{\mathbf{k}\alpha} \omega, \quad \mathbf{P} = \sum_{\mathbf{k}\alpha} N_{\mathbf{k}\alpha} \mathbf{k}. \quad (3.1)$$

These formulae make it possible to introduce the concept fundamental to all of quantum electrodynamics: that of *light quanta*, or *photons*³. Namely, the free electromagnetic field can be regarded as a collection of particles, each carrying energy $\omega (= \hbar\omega)$ and momentum $\mathbf{k} (= \hbar\omega/c)$. The relationship between a photon’s energy and momentum is precisely what relativistic mechanics requires for particles of zero rest mass that propagate at the speed of light. The occupation numbers $N_{\mathbf{k}\alpha}$ acquire the meaning of the number of photons with given momenta $\mathbf{k}$ and polarizations $\mathbf{e}^{(\alpha)}$. The polarization property of a photon is analogous to the notion of spin for other particles (the specific features of the photon in this regard will be examined below, in § 6).

It is easy to see that the mathematical formalism developed in the preceding section is fully consistent with the picture of the electromagnetic field as an assembly of photons; it is nothing other than the apparatus of so-called second quantization as applied to a system of photons⁴. In this method (see Vol. III, § 64) the occupation numbers serve as the independent variables, and the operators act on functions of those numbers. The central role is played by the “annihilation” and “creation” operators for particles, which respectively decrease or increase an occupation number by one. Exactly such operators are $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}_{\mathbf{k}\alpha}^+$: the operator $\hat{c}_{\mathbf{k}\alpha}$ destroys a photon in the state $\mathbf{k}\alpha$, while $\hat{c}_{\mathbf{k}\alpha}^+$ creates one in that state.

The commutation rule (2.16) corresponds to particles obeying Bose statistics. Photons are therefore bosons, as one might have anticipated: the number of photons occupying any given state can be arbitrary (we shall return to the significance of this fact in § 5).

The plane waves $\mathbf{A}_{\mathbf{k}\alpha}$ (2.26), which appear in the operator $\hat{\mathbf{A}}$ (2.17) as coefficients multiplying the photon annihilation operators, can be interpreted as the wave functions of photons possessing definite momenta $\mathbf{k}$ and polarizations $\mathbf{e}^{(\alpha)}$. This interpretation corresponds to expanding the ψ -operator as a series in the stationary-state wave functions of the particle within the non-relativistic second-quantization framework (unlike the

²This subtraction can be carried out in a formally consistent manner by agreeing to interpret the operator products in (2.10) as “normal-ordered”, i.e. arranged so that the operators $\hat{c}^+$ always stand to the left of the operators $\hat{c}$. Formula (2.23) then takes the form

$$\hat{H} = \sum_{\mathbf{k}\alpha} (\omega \hat{c}_{\mathbf{k}\alpha}^+ \hat{c}_{\mathbf{k}\alpha}).$$

³The photon concept was first put forward by *Einstein* (A. Einstein, 1905).

⁴The method of second quantization as applied to radiation theory was developed by *Dirac* (P. A. M. Dirac, 1927).

non-relativistic case, however, the expansion (2.17) involves both annihilation and creation operators; the significance of this distinction will become clear later, see § 12).

The wave function (2.26) is normalized by the condition

$$\int \frac{1}{4\pi} (|\mathbf{E}_{\mathbf{k}\alpha}|^2 + |\mathbf{H}_{\mathbf{k}\alpha}|^2) d^3x = \omega \quad (3.2)$$

This amounts to normalizing to one photon in a volume $V = 1$. Indeed, the integral on the left-hand side of the equation represents the quantum-mechanical expectation value of the photon's energy in a state described by the given wave function⁵. On the right-hand side of (3.2) stands the energy of a single photon.

The role of the “Schrödinger equation” for the photon is played by Maxwell's equations. In the present case (for the potential $\mathbf{A}(\mathbf{r}, t)$ satisfying condition (2.1)) this is the wave equation

$$\frac{\partial^2 \mathbf{A}}{\partial t^2} - \Delta \mathbf{A} = 0.$$

In the general case of arbitrary stationary states, the photon “wave functions” are complex solutions of this equation whose time dependence enters through the factor $e^{-i\omega t}$.

Speaking of the photon wave function, let us emphasize once more that it cannot by any means be treated as a probability amplitude for the spatial localization of a photon — in contrast to the basic meaning of the wave function in non-relativistic quantum mechanics. This is linked to the fact that (as noted in § 1) the notion of photon coordinates has no physical meaning at all. We shall revisit the mathematical side of this issue at the end of the next section.

The Fourier-transform components of $\mathbf{A}(\mathbf{r}, t)$ with respect to the coordinates constitute the photon wave function in the momentum representation; we denote it $\mathbf{A}(\mathbf{k}, t) = \mathbf{A}(\mathbf{k})e^{-i\omega t}$. Thus, for a state with a definite momentum $\mathbf{k}$ and polarization $\mathbf{e}^{(\alpha)}$ the momentum-representation wave function is given simply by the coefficient of the exponential factor in (2.26):

$$\mathbf{A}_{\mathbf{k}\alpha}(\mathbf{k}', \alpha') = \sqrt{4\pi} \frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}} \delta_{\mathbf{k}'\mathbf{k}} \delta_{\alpha\alpha'}. \quad (3.3)$$

In keeping with the measurability of the momentum of a free particle, the momentum-representation wave function carries deeper physical content than its coordinate-space counterpart: it provides a way to compute the probabilities $w_{\mathbf{k}\alpha}$ of the various values of momentum and polarization for a photon in a given state. According to the general rules of quantum mechanics, $w_{\mathbf{k}\alpha}$ is given by the squared modulus of the expansion coefficients of $\mathbf{A}(\mathbf{k}')$ in the wave functions of states with definite $\mathbf{k}$ and $\mathbf{e}^{(\alpha)}$:

$$w_{\mathbf{k}\alpha} \propto \left| \sum_{\mathbf{k}'\alpha'} \mathbf{A}_{\mathbf{k}\alpha}^*(\mathbf{k}', \alpha') \mathbf{A}(\mathbf{k}') \right|^2$$

⁵Note that the coefficient $1/(4\pi)$ in the integral (3.2) is twice the usual coefficient $1/(8\pi)$ in (2.10). This difference is ultimately connected with the complex character of the vectors $\mathbf{E}_{\mathbf{k}\alpha}$, $\mathbf{H}_{\mathbf{k}\alpha}$, as opposed to the Hermitian field operators $\hat{\mathbf{E}}$, $\hat{\mathbf{H}}$.

(the proportionality coefficient depends on the choice of normalization for the functions). Substituting (3.3), we obtain

$$w_{\mathbf{k}\alpha} \propto |\mathbf{e}^{(\alpha)} \mathbf{A}(\mathbf{k})|^2. \quad (3.4)$$

Summing over the two polarizations, we find the probability that a photon has momentum $\mathbf{k}$:

$$w_{\mathbf{k}} \propto |\mathbf{A}(\mathbf{k})|^2. \quad (3.5)$$

§4. Gauge invariance

As is well known, the choice of field potentials in classical electrodynamics is not unique: the components of the 4-potential A_μ may be subjected to an arbitrary *gauge* (or *gradient*) transformation of the form

$$A_\mu \rightarrow A_\mu + \partial_\mu \chi, \quad (4.1)$$

where χ is an arbitrary function of the coordinates and time (see Vol. II, § 18). In the case of a plane wave, restricting oneself to transformations that preserve the functional form of the potential (its proportionality to $\exp(-ik_\mu x^\mu)$), the non-uniqueness amounts to the freedom of adding to the wave amplitude an arbitrary 4-vector proportional to k^μ .

The ambiguity of the potential persists, of course, in the quantum theory as well — as applied to the field operators or to the photon wave functions. Without prejudging the choice of potentials, one should write in place of (2.17) an analogous expansion for the operator 4-potential

$$\hat{A}^\mu = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} A_{\mathbf{k}\alpha}^\mu + \hat{c}_{\mathbf{k}\alpha}^+ A_{\mathbf{k}\alpha}^{\mu*}), \quad (4.2)$$

where the wave functions $A_{\mathbf{k}\alpha}^\mu$ are 4-vectors of the form

$$A_{\mathbf{k}}^\mu = \sqrt{4\pi} \frac{e^\mu}{\sqrt{2\omega}} e^{-ik_\nu x^\nu}, \quad e_\mu e^{\mu*} = -1,$$

or, in a compact notation that suppresses the four-dimensional vector indices:

$$A_{\mathbf{k}} = \sqrt{4\pi} \frac{e}{\sqrt{2\omega}} e^{-ikx}, \quad ee^* = -1. \quad (4.3)$$

Here the 4-momentum is $k_\mu = (\omega, \mathbf{k})$ (so that $kx = \omega t - \mathbf{k}\mathbf{r}$), and e is the unit polarization 4-vector⁶.

If one restricts oneself to gauge transformations that do not alter how the function (4.3) depends on the coordinates and time, they amount to the replacement

$$e_\mu \rightarrow e_\mu + \chi k_\mu, \quad (4.4)$$

⁶Expression (4.3) does not have an entirely relativistically covariant (4-vector) form, which is related to the non-invariant character of our normalization to a finite volume $V = 1$. This is, however, of no fundamental significance and is fully compensated by the convenience of such a normalization scheme. We shall see later that it ensures simple and automatic derivation of real physical quantities in the proper invariant form.

where $\chi = \chi(k^\mu)$ is an arbitrary function. The transversality of the polarization means that one can always choose a gauge in which the 4-vector e takes the form

$$e^\mu = (0, \mathbf{e}), \quad \mathbf{e}\mathbf{k} = 0 \quad (4.5)$$

(we shall call such a gauge *three-dimensionally transverse*). In an invariant four-dimensional notation this requirement is written as the condition of *four-dimensional transversality*

$$ek = 0. \quad (4.6)$$

Note that this condition (as well as the normalization condition $ee^* = -1$) is not violated by the transformation (4.4), by virtue of $k^2 = 0$. On the other hand, the vanishing of the 4-momentum squared of a particle signifies that its mass is zero. A direct link between gauge invariance and the vanishing of the photon mass is thereby revealed (further aspects of this connection will be discussed in § 14).

No measurable physical quantity should change under a gauge transformation of the wave functions of the photons taking part in a given process. This requirement of *gauge invariance* plays an even greater role in quantum electrodynamics than in the classical theory. As we shall see from numerous examples, it serves here, alongside the requirement of relativistic invariance, as a powerful heuristic principle.

In turn, the gauge invariance of the theory is closely bound up with the law of conservation of electric charge; we shall dwell on this aspect in § 43.

We already remarked in the preceding section that the coordinate-space wave function of a photon cannot be interpreted as a probability amplitude for its spatial localization. From a mathematical standpoint this circumstance manifests itself in the impossibility of constructing, out of the wave function, a quantity whose formal properties alone would qualify it to serve as a probability density. Such a quantity would have to be an essentially positive bilinear combination of the wave function A_μ and its complex conjugate. Moreover, it would need to satisfy definite requirements of relativistic covariance — namely, to be the time component of a 4-vector (the point being that the continuity equation expressing conservation of particle number takes the four-dimensional form of a vanishing 4-divergence of the current 4-vector; the time component of this current is, in the present context, the probability density for the localization of the particle; see Vol. II, § 29). On the other hand, gauge invariance demands that A_μ enter the current only through the antisymmetric tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = -i(k_\mu A_\nu - k_\nu A_\mu)$. The current 4-vector would therefore have to be assembled bilinearly from $F_{\mu\nu}$ and $F_{\mu\nu}^*$ (together with components of the 4-vector k_μ). But no such 4-vector can be constructed at all: every expression satisfying the stated conditions (for instance, $k^\lambda F_{\mu\nu}^* F_\lambda{}^\nu$) vanishes by the transversality condition ($k^\lambda F_{\nu\lambda} = 0$), quite apart from the fact that it would not be essentially positive (since it contains odd powers of the components k_μ).

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